# Theeraj Chandra

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## Education

**University of Waterloo** Sep 2022 – Apr 2027
*Bachelor of Applied Sciences in Computer Engineering*

- **Relevant courses:** Distributed Computing, Intro Machine Learning, Real-time OS, Databases, Computer Networks

## Technical Skills

**Languages**: Rust, Python, Ruby, C, C++, Go, Java, GraphQL, TypeScript, Verilog
**Libraries & Frameworks**: Ruby on Rails, React, PyTorch, Angular, Next.js
**Tools & Databases**: Buildkite, Grafana, Docker, Sentry, Redis, PostgreSQL

## Experience

**Software Engineer Intern (QE Infra)** | *TypeScript, GraphQL, Go, Buildkite* Jan 2026 – Apr 2026
*StackAdapt* *Toronto, ON*

- Optimized high-throughput analytical queries for a 14GB PostgreSQL dataset, achieving a **42x performance speedup** (5m to 7s) by re-architecting composite indices and resolving data non-determinism caused by timestamp collisions.
- Authored the design specification for an **AI-powered Triage Engine** utilizing a headless MCP architecture and temporal knowledge graphs to automate failure root-cause analysis across distributed CI pipelines.
- Engineered a high-concurrency **TypeScript Service** that scrapes historical data from 500+ Buildkite streams; built this data-aggregator to feed LLMs the context needed to accurately distinguish between real bugs and environmental flakiness.
- Optimized CI throughput by building a custom **workload scheduler** in **Go** to dynamically shard test suites across distributed clusters; reduced the feedback loop for our E2E pipelines from 40 minutes to **under 8 minutes.**

**Software Engineer Intern** | *React.js, TypeScript, Material UI* Jan 2025 – Apr 2025
*SuperWorld Inc.* *Miami, FL*

- Engineered a geolocation-based **virtual check-in** feature for SuperWorld Map using the Haversine formula algorithm, enabling users to authenticate location proximity within a 300m radius, enhancing user engagement through location verification.
- Played a key role in developing a proprietary **conversational AI interface** powered by Meta LLaMA 3, enhancing user retention through personalized map interactions and intuitive mobile-first UX.
- Built a React + TypeScript dashboard with async parallel API calls for real-time KPI updates and interactive Recharts visualizations.

**Test Engineer Intern** | *Azure DevOps, Perl, Postman* Jan 2024 – Aug 2024
*First National Financial LP* *Toronto, ON*

- Developed **Python** and **Perl** scripts to automate pooling target validation and improve testing processes, ensuring higher reliability and reducing manual efforts by **30%.**
- Tested mortgage broker tools for BMO, TD, and Manulife on staging and QA environments, performing **regression**, **UAT**, and **exploratory** tests to validate functionality and client-specific requirements.

## Projects

**Distributed ML Training Runtime** | *Rust, Python*

- Built a **fault-tolerant distributed runtime** in **Rust** and **Python**, coordinating sharded data loading and checkpointing across **async** concurrent workers via **gRPC** and **tokio**; with automatic shard rebalancing and **PyO3** bindings for a PyTorch-compatible API.
- Implemented a two-phase **commit checkpoint protocol** with atomic writes to **S3-compatible** object storage and end-to-end **Prometheus** observability, ensuring zero data corruption across fault injection.

**Zero-Allocation Limit Order Book** | *Rust*

- Engineered a matching engine from scratch in Rust achieving **160ns** average latency and **6.2M ops/sec**, outperforming typical production exchange requirements by an order of magnitude.
- Architected a **slab/arena** allocator and intrusive data structures, eliminating heap churn and lock contention to ensure deterministic $O(1)$ allocation and cache-friendly $O(\log P)$ matching.